The Perils of Bilateral Sovereign Debt

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Official Sovereign Debt

- A large share of sovereign borrowing takes the form of official debt
  - … Multilaterals, development banks, other governments

- Emergence of new bilateral creditors **outside** the Paris Club IDS data
  - … with claims to seniority and sometimes **confidential** terms

Questions

- How does the presence of a large senior lender affect sovereign debt markets?
- What are its welfare implications for borrowing governments?

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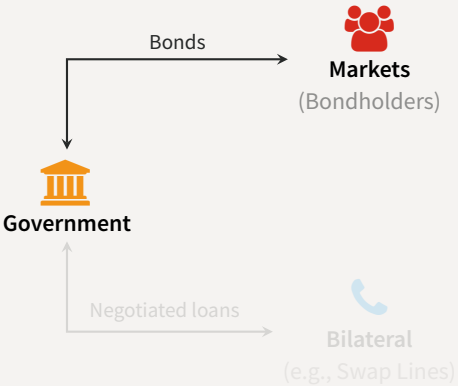
Evaluating Senior Official Creditors

Quantitative sovereign debt model with

- Competitive creditors in private markets
- Large bilateral lender
  1. Superior enforcement
     [de-facto seniority]
  2. Bargained terms
     [price and quantity]
  3. Short-maturity loans
- Prime example: Central Bank swap lines
  (Horn et al., 2021)
- Focus on the interaction between both funding sources



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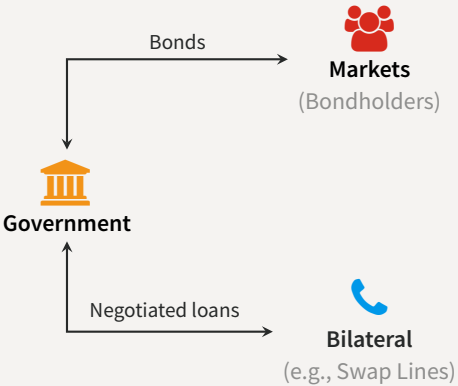

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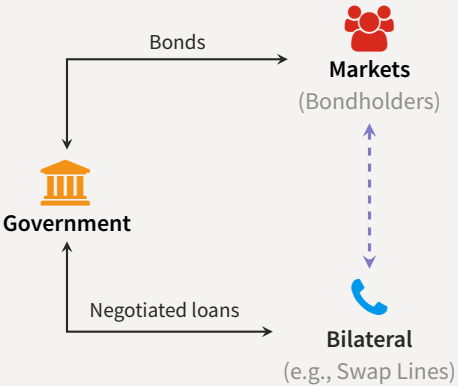

Relational Overborrowing

Main findings

- Bilateral loans have significant effects on equilibrium outcomes
  - … provide funding when other sources dry up (e.g. because of default risk) ▲
  - … can also incentivize more **risk-taking** ▼
- If the rate on bilateral loans is decreasing in *market* debt [cross-elasticity]
  - … government issues market debt more quickly, delevers more slowly
  - … spends longer in the risky region
  - … defaults more frequently
  - … welfare losses for the government
- Cross-elasticity can emerge endogenously from bargaining
  - … at plausible values for bargaining weights
  - … increased frequency of defaults dominates extra liquidity
  - … **relational overborrowing**

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Literature

- Sovereign debt/default with interactions from ‘official’ debt
  - … senior debt (Hatchondo, Martinez & Önder 2017), senior debt with conditionality (Boz 2011, Fink & Scholl 2016), bailout agencies (Corsetti, Guimarães & Roubini 2006, Kirsch & Rühmkorf 2017, Roch & Uhlig 2018), official debt (Arellano & Barreto 2024, Liu, Liu & Yue 2025)
- Data on new official creditors
  - … Horn, Reinhart & Trebesch 2021a, 2021b, Gelpern et al. 2021, Horn, Parks, Reinhart & Trebesch 2023
- Central Bank swap lines
  - … among advanced economies (Bahaj & Reis 2021, Cesa-Bianchi, Eguren-Martin & Ferrero 2022), data for emerging-market borrowers (Perks, Rao, Shin & Tokuoka 2021)

Roadmap

- Theory
  - Bargaining and the Cross-Elasticity
  - The Cross-Elasticity and Debt Dilution
- Quantitative Model
  - Endogenous Bargaining
  - Programming the Large Lender
- Concluding remarks

Theory

Theory

Bargaining and the Cross-Elasticity

Bargaining Induces a Cross-Elasticity

- Two periods $t \in \{0, 1\}$, endowments $y_0 < y_1$
- Government discount factor β, lender's discount factor $\beta_L > \beta$
- **Bargaining** transfer x at $t = 0$ for repayment $x(1 + r)$ at $t = 1$, weight θ for lender

$$\max_{x,r} \mathcal{L}(x, r)^{\theta} \times \mathcal{B}(x, r; y_0, y_1)^{1-\theta}$$

where

$$\mathcal{L}(x, r) = \underbrace{-x + \beta_L x(1 + r)}_{\text{agreement}}$$

$$\mathcal{B}(x, r; y_0, y_1) = \underbrace{u(y_0 + x) + \beta u\left(y_1 - x(1 + r)\right)}_{\text{agreement}} - \underbrace{\left(u(y_0) + \beta u\left(y_1\right)\right)}_{\text{threat point}}$$

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Bargaining Induces a Cross-Elasticity

- First-order conditions for bargaining problem

$$u'(y_0 + x) = \frac{\theta}{1-\theta} \frac{\mathcal{B}(x, r; y_0, y_1)}{\mathcal{L}(x, r)}$$

Surplus sharing

$$u'(y_0 + x) = \frac{\beta}{\beta_L} u'\left(y_1 - x(1 + r)\right)$$

Euler

- Can prove that

$$\frac{\partial x}{\partial y_0} \leq 0 \qquad \text{and} \qquad \frac{\partial r}{\partial y_0} \leq 0$$

- **Later** y_0, y_1 are given by market debt dynamics, so
  - Recent issuances: increases y_0 and decreases y_1, makes you stronger
  - Legacy debt: lowers y_0 and y_1, typically makes you weaker

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Theory

The Cross-Elasticity and Debt Dilution

Debt Dilution with Undefaultable Loans

Issue long bonds b at $t = 0$, short bonds d and loans m at $t = 1$, repay at $t = 2$

$$c_0 = q(b+d)b \qquad c_1 = q(b+d)d + \phi(b,d,m)m \qquad c_2(z) = y(z) - m - \min\{h(z), b+d\}$$

with

$$q(b+d) = \mathbb{P}\left(b + d \leq h(z)\right) \qquad \text{and} \qquad m \leq \bar{m}$$

to maximize $\sum_{t=0}^{2} \mathbb{E}\left[u(c_t)\right]$

Commitment: Choose d internalizing effect on initial prices

$$u'(c_0)\underbrace{q'(b+d)b}_{\text{past prices}} + u'(c_1)\Big(\underbrace{q(b+d)}_{\text{revenue}} + \underbrace{q'(b+d)d}_{\text{current prices}} + \underbrace{\phi'_d(b,d,m)m}_{\text{cross-elasticity}}\Big) = q(b+d)\mathbb{E}\left[u'(c_2^R)\right]$$

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not under discretion

Debt Dilution with Undefaultable Loans and Cross-Elasticity

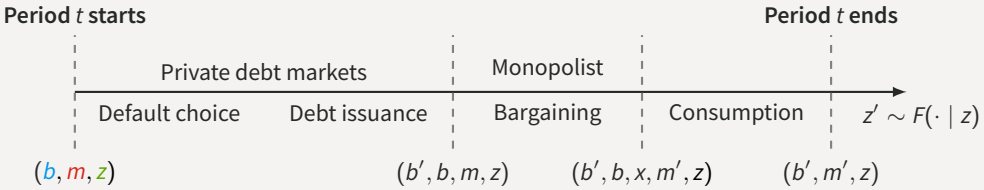
Takeaways

- Bargaining can induce a cross-elasticity from market debt to loan terms
  … through threat-point manipulation
- The cross-elasticity worsens the debt dilution problem
  … it can also create welfare losses if strong enough

Quantitative Model

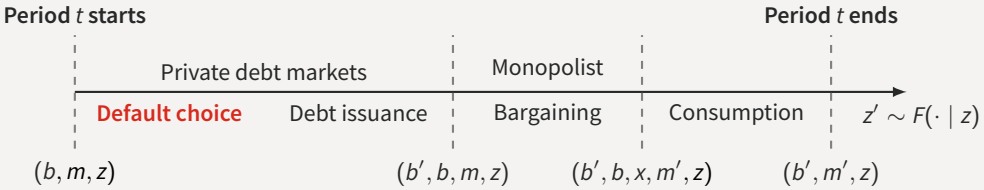
Timeline of Events

- Enter period t owing b to bondholders, m to monopolist, income $y(z)$



Timeline of Events

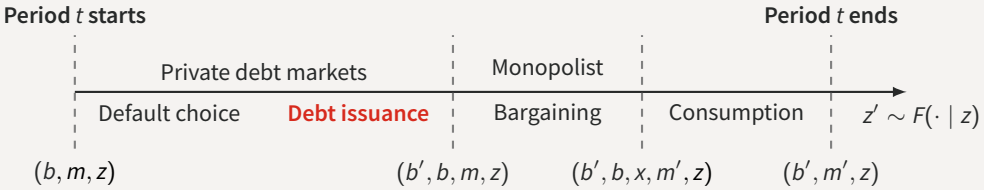
- Choose to repay or default the *market* debt subject to convex output costs



$$v(b, m, z) = \max\left\{v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D\right\}$$

Timeline of Events

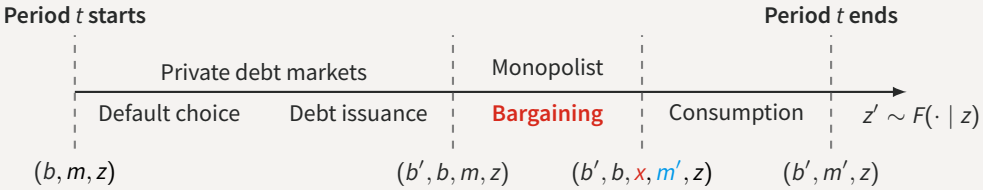
- If repaid, issue new (long-term) debt b' in markets at price q [coupon rate κ, maturity $1/\delta$]



$$q(b', b, m, z) = \beta_L \mathbb{E}\left[(1 - 1_{\mathcal{D}}(b', m', z'))\,(\kappa + (1-\delta)q(b'', b', m', z')) \mid z\right]$$

Timeline of Events

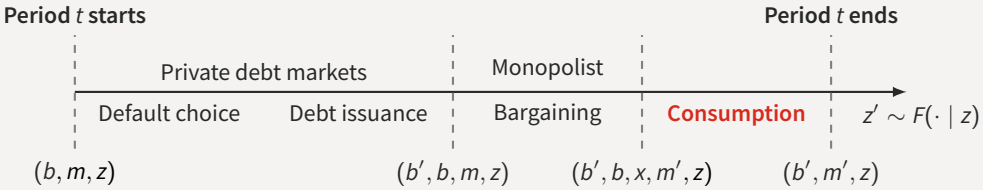
- Meet with senior lender, decide any transfers x and new/remaining balance m'



$$r(x, m', m) = \frac{m'}{x + m} - 1$$

Timeline of Events

- Consume output plus revenues from debt issuance plus transfers minus debt service



$$c_R = y(z) + q(b', b, m, z)(b' - (1-\delta)b) + x_R(b', b, m, z) - \kappa b$$

$$c_D = y(z) - h(z) + x_D(m, z); \qquad x_D(m, z) \leq \Gamma(m)$$

Calibration

- Standard strategy in sovereign debt: match Argentina 1993-2001 to model w/o swaps

	Parameter	Value
Sov's discount factor	β	0.9605
Sov's risk aversion	γ	2
Pref. scale	χ	0.025
Lender's barg. power	θ	0.5
Risk-free rate	r	0.01
Duration of debt	ρ	0.05
Income autocorr.	ρ_z	0.9484
Std. dev. y_t	σ_z	0.02
Reentry prob.	ψ	0.0385
Default cost: linear	d_0	-0.2466
Default cost: quadratic	d_1	0.2922

	Data	Only market
Avg spread (bps)	815	796
Std spread (bps)	443	499
Debt-to-GDP (%)	17.4	18

Note moments from pre-default samples

Quantitative Model

Endogenous Bargaining

Bargaining Stage with Monopolist

- At state z, owing debt b bonds and m on the loan and having issued b'

Government surplus

$$\max_{x,m'} \mathcal{L}_R(b', x, m, m', z)^{\theta} \times \mathcal{B}_R(b', b, x, m, m', z)^{1-\theta}$$

Lender surplus

- Lender's surplus

$$\mathcal{L}_R(b', x, m, m', z) = \underbrace{\left(-x + \beta_L \mathbb{E}\left[h(b', m', z') \mid z\right]\right)}_{\text{agreement}} - \underbrace{\left(m + \beta_L \mathbb{E}\left[h(b', 0, z') \mid z\right]\right)}_{\text{threat point}}$$

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$$\mathcal{B}_R(b', b, x, m, m', z) = \underbrace{u\big(y(z) + B(b', b, m, z) + x\big) + \beta \mathbb{E}\left[v(b', m', z') \mid z\right]}_{\text{agreement}}$$
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same sdf as markets

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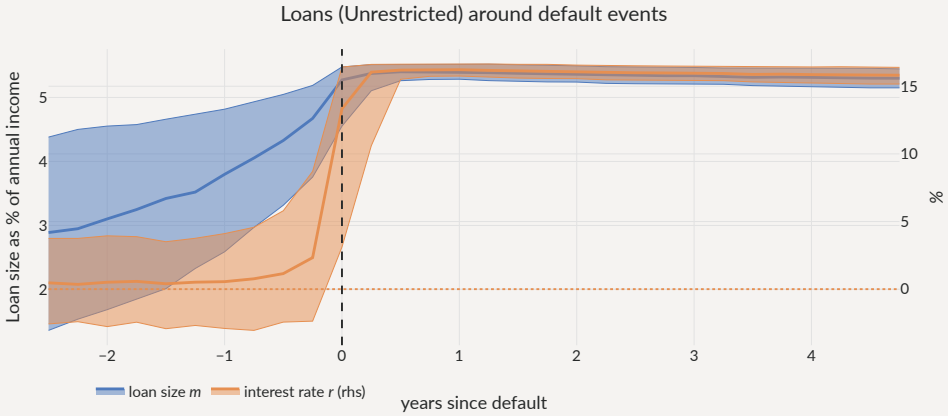
Usage of Loans

- Sharp acceleration before defaults, maxed out *in* default if possible



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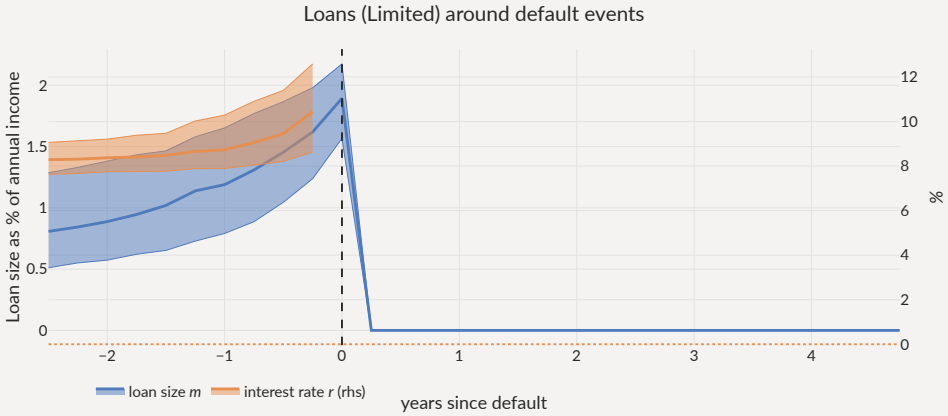

Limiting Loans in Default

- **Limited**: entire loan must be repaid while in default $\Gamma(m) = 0$

	Only market	Unrestricted, $\theta = 0.5$	Limited, $\theta = 0.5$
Avg spread (bps)	682	1,654	866
Std spread (bps)	469	1,248	599
$\sigma(c)/\sigma(y)$ (%)	108	104	109
Debt to GDP (%)	15.9	16.2	16.5
Loan to GDP (%)	0	2.46	0.863
Loan spread (bps)	–	-318	441
Corr. loan & spreads (%)	–	68.2	69.1
Default frequency (%)	5.42	10.9	6.62
Welfare gains (rep)	–	-0.052%	0.034%

Note Statistics from each ergodic distribution

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- Default rates: ↑ with loans (less with Limited)

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- Welfare: $\approx$ with loans (Limited $\succcurlyeq$ Unrestricted)

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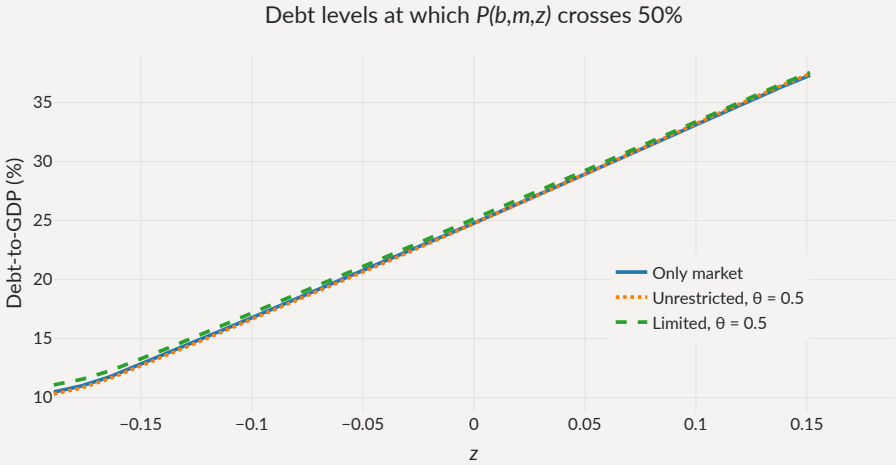
Avg spread (bps)	682	1,654	866
Std spread (bps)	469	1,248	599
$\sigma(c)/\sigma(y)$ (%)	108	104	109
Debt to GDP (%)	15.9	16.2	16.5
Loan to GDP (%)	0	2.46	0.863
Loan spread (bps)	–	-318	441
Corr. loan & spreads (%)	–	68.2	69.1
Default frequency (%)	5.42	10.9	6.62
Welfare gains (rep)	–	-0.052%	0.034%

Note Statistics from each ergodic distribution

Default Barriers with Loans

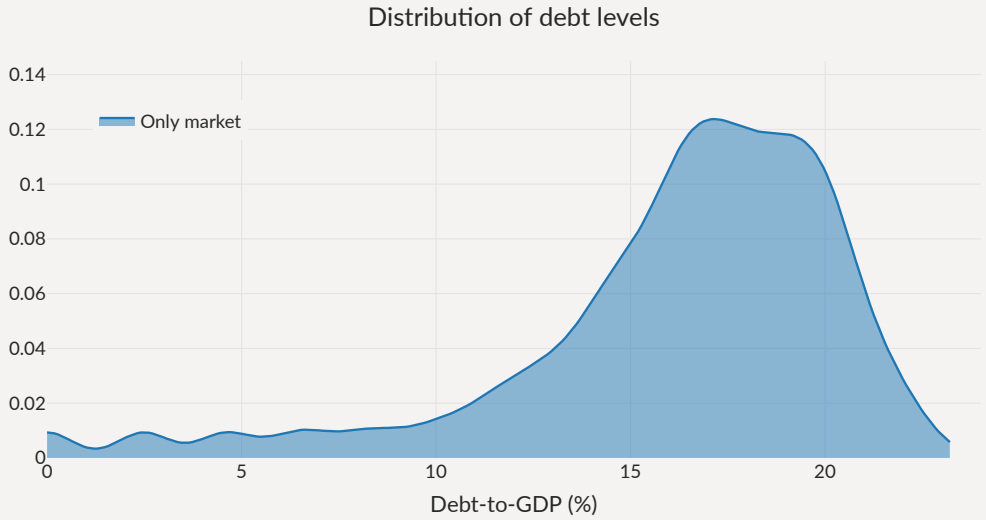
- Both Unrestricted and Limited: default barrier marginally outward



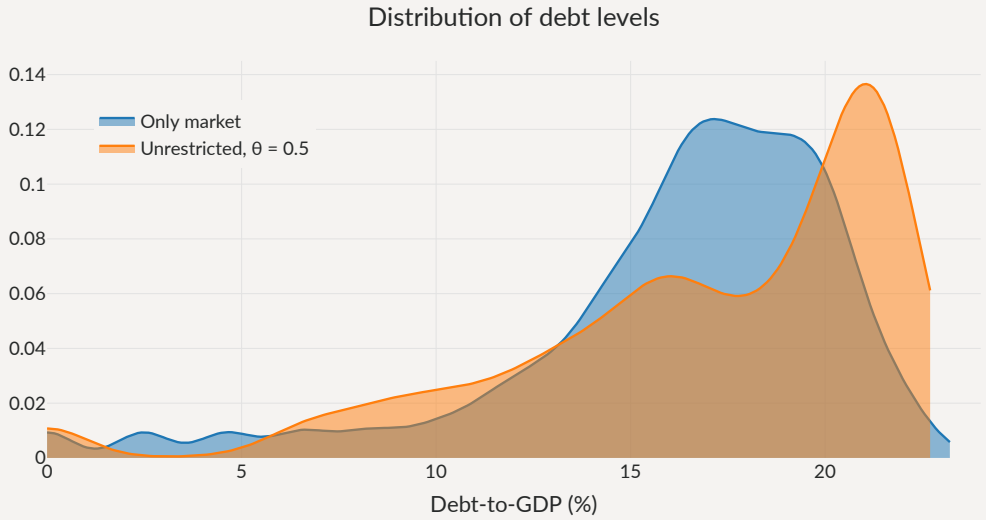
If Loans help repay the debt,

Why are there more defaults with loans?

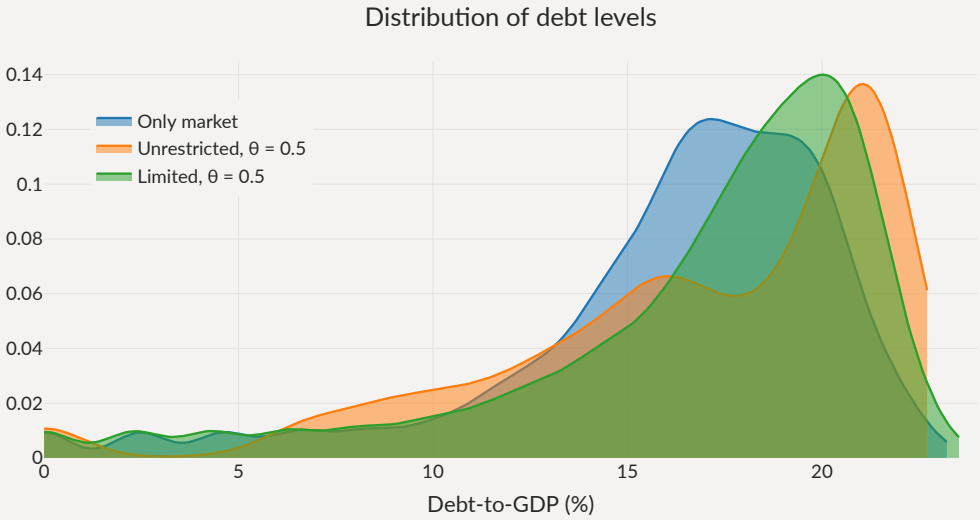
Debt Levels with Loans



Debt Levels with Loans



Debt Levels with Loans



Debt Prices with Loans

Lower prices with same ex-post default rates: relational overborrowing similar to debt dilution

Relational Overborrowing

Government's surplus

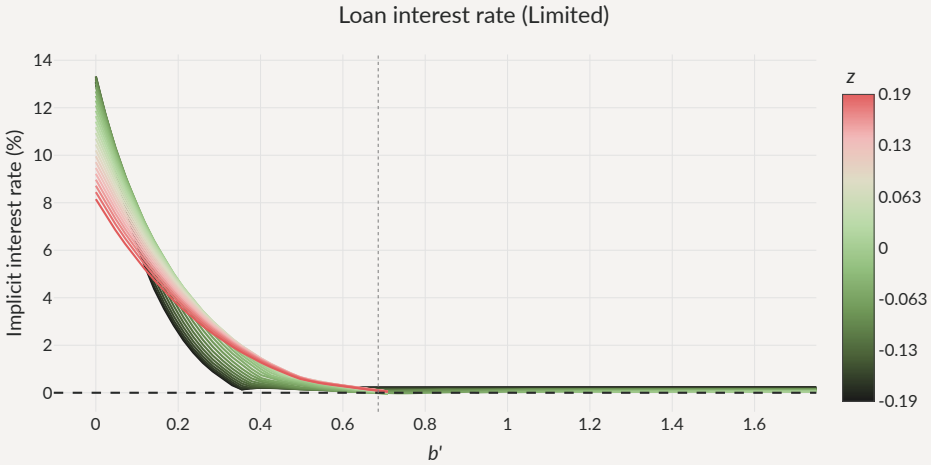
$$\mathcal{B}_R(b', b, x, m, m', z) = u\big(y(z) + B(b', b, m, z) + x\big) + \beta\mathbb{E}\left[v(b', m', z') \mid z\right]$$
$$- \big(u\big(y(z) + B(b', b, m, z) - m\big) + \beta\mathbb{E}\left[v(b', 0, z') \mid z\right]\big)$$

- Revenues from debt issuance $B(b', b, m, z)$ modulate the value of the threat point
  - After large revenues (high q, high b'), gov't flush with cash, **strong** in bargaining
  - After bad issuance (low q *or* low b'), gov't **weak** in bargaining

Risk-taking Incentives

Drawings

Threat point manipulation: increase market b' to reduce bilateral r **Cross-elasticity**



Relational Overborrowing and the Cross-elasticity

Government's Euler equation for bonds

$$u'(c)\left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r}\frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r_b}}{\partial b'} m'\right) = \beta \mathbb{E}\left[v_b(b', m', z') + v_m(b', m', z')\frac{\partial m'}{\partial b'} \mid z\right]$$

- An extra bond:
  - … yields marginal revenue $q + \frac{\partial q}{\partial b'} i$ [standard debt-Laffer curve effect]
  - … makes you **stronger** in bargaining today
    - … affects the amount m' borrowed from monopolist
    - … affects terms of borrowing with monopolist
  - … makes you **weaker** in bargaining tomorrow

Relational Overborrowing and the Cross-elasticity

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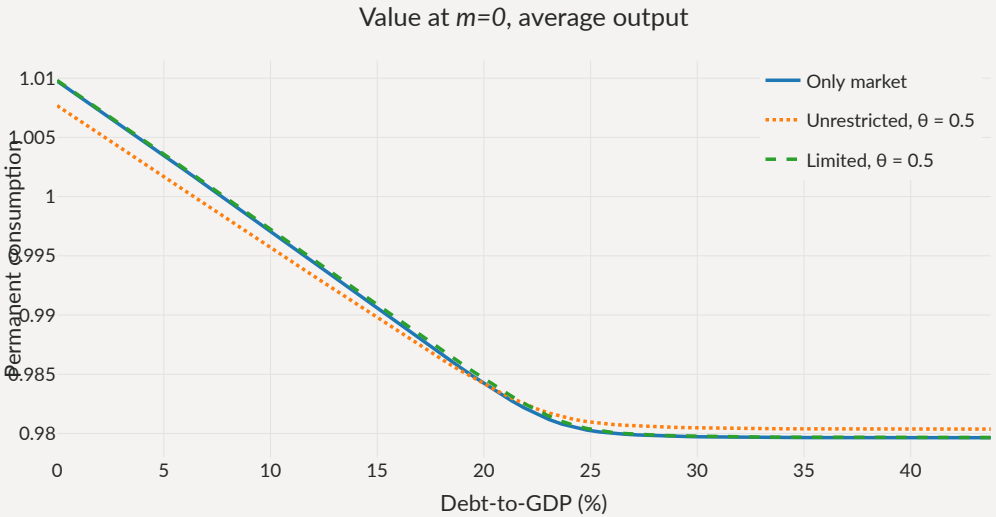
Welfare Effects of Bilateral Loans

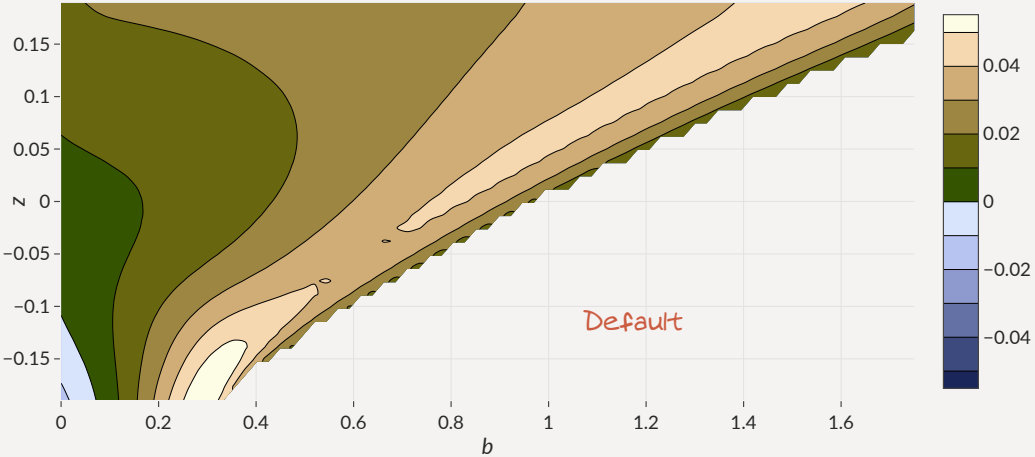
Limited $\succcurlyeq$ Unrestricted, but…



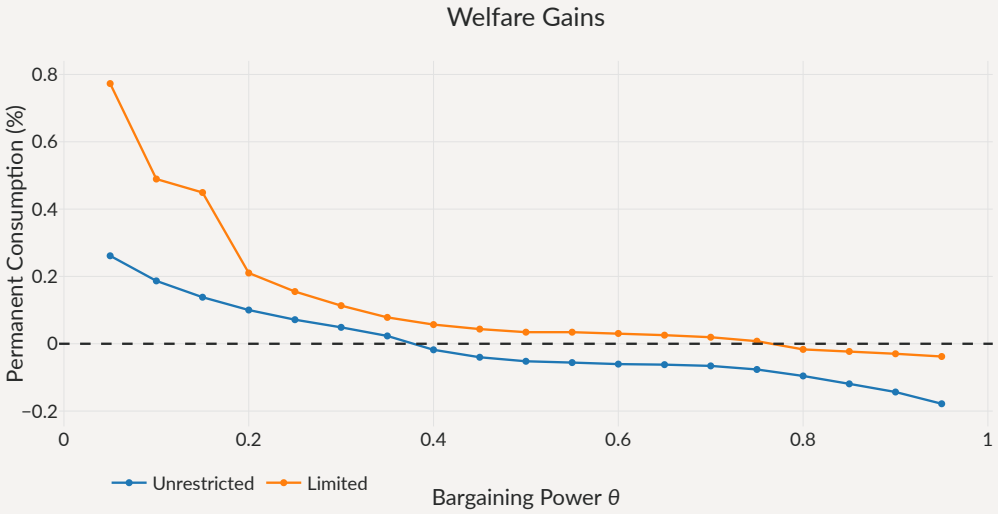
Welfare Effects of Bilateral Loans (cont'd)

Overall welfare effect small: more risk, but more frontloading in short run.

v(b,m=0,z) - v(b,z)



Bargaining Power and Welfare



Quantitative Model

Programming the Large Lender

Possible Rules

- Explore interest rate rules of the form

$$r(b', m') = \max\{r^\star, \alpha_0 + \alpha_b b' + \alpha_m m'\}$$

- Two versions

Size-dependent

$\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$

Risk-inducing

$\alpha_0 > 0, \alpha_b < 0, \alpha_m \geq 0$

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Equilibrium with Exogenous Rules

- 'Only market' standard calibration to Argentina 1993-2001

	Only market	Size dependent r	Risk inducing r	Limited, $\theta = 0.5$
Avg spread (bps)	690	615	908	878
Std spread (bps)	513	453	798	647
$\sigma(c)/\sigma(y)$ (%)	103	103	106	104
Debt to GDP (%)	16.2	16.5	16.6	16.8
Loan to GDP (%)	0	0.322	0.792	0.877
Loan spread (bps)	–	660	3,488	444
Corr. loan & spreads (%)	–	56.2	38.5	64.1
Default frequency (%)	5.44	4.95	6.51	6.63
Welfare gains (rep)	–	0.13%	-0.021%	0.034%

Equilibrium with Exogenous Rules

- Default rates: ↓ with size dependent ↑ with risk-inducing

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Equilibrium with Exogenous Rules

- Spreads: ↓ with size dependent ↑ with risk-inducing

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Equilibrium with Exogenous Rules

- Welfare: ↑ with size dependent ↓ with risk-inducing

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Concluding remarks

The Perils of Bilateral Sovereign Debt

- Simple model of borrowing from markets and a senior bilateral lender
  - … strong interaction between two markets for sovereign debt
  - … even if bilateral loans are **not** used intensely on the equilibrium path

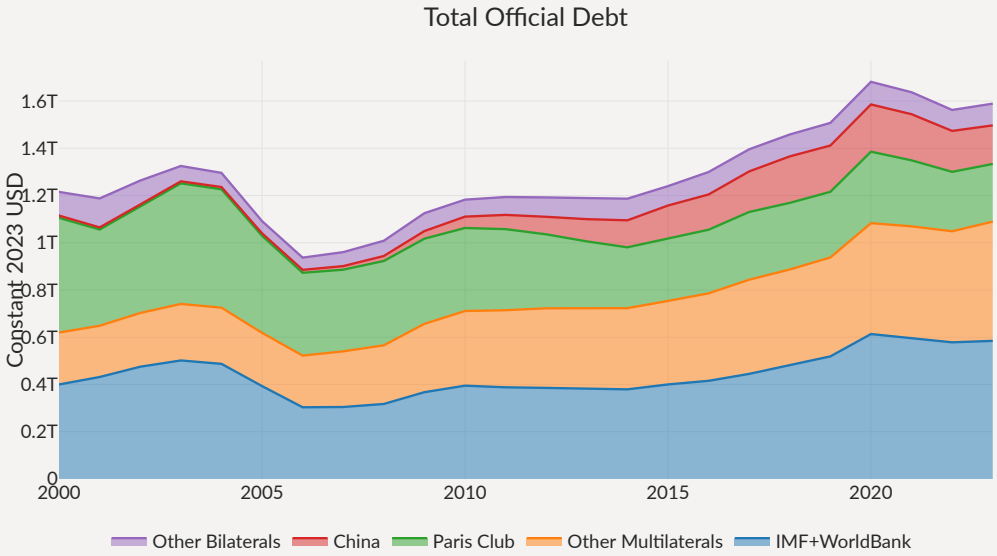
- **Dangerous** when bilateral interest rate responds negatively to *market* debt
  - … cross-elasticity induces risk-taking, more defaults, welfare losses
  - … Bargaining as an example of situation where cross-elasticity emerges

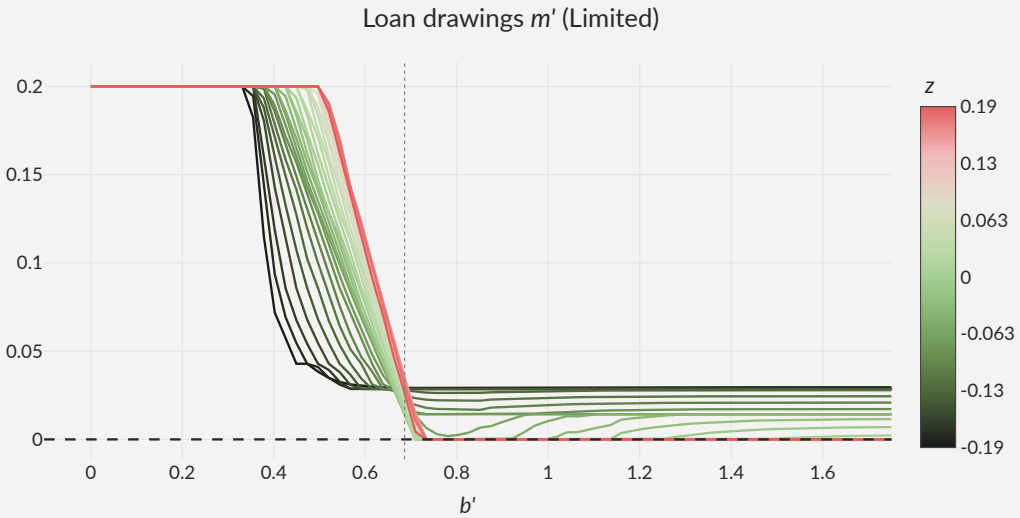
- Cross-elasticity constitutes a simple test to assess welfare gains of **new** instruments
  - … or a boost to the gains of fiscal rules, state-contingent debt…

Scan to find the paper

A New Landscape for Official Sovereign Debt

Back



Risk-taking Incentives

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